

Emergency Airway Management in Penetrating Neck Injury

From the Department of Emergency Medicine,* University of Southern California School of Medicine, and the Department of Emergency Medicine,† Los Angeles County–University of Southern California Medical Center, Los Angeles, CA.

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Address for reprints:
Diku Mandavia, MD, FRCPC,
Department of Emergency Medicine,
Room 1011, General Hospital, Los
Angeles County–University of
Southern California Medical Center,
1200 North State Street, Los Angeles,
CA 90033; 323-226-6692,
fax 323-226-6454;
E-mail mandavia@usc.edu.

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Diku P. Mandavia, MD**
Stephanie Qualls, MD**
Ivan Rokos, MD†

Study objectives: Airway management in the context of penetrating neck injury is a challenging scenario. Management decisionmaking has not been well studied and the initial airway approach remains controversial. We examined various initial emergency airway techniques and their success in the setting of penetrating neck trauma.

Methods: A retrospective study was conducted of emergency department intubations in penetrating neck injury from January 1, 1993, to December 31, 1996, at a Level I trauma center. Cases of out-of-hospital traumatic arrest or out-of-hospital intubation were excluded. Successful airway management was defined as endotracheal tube placement confirmed by clinical evaluation, pulse oximetry, chest radiography, and end-tidal CO₂ detection.

Results: During the study period, 748 consecutive patients with penetrating neck injury were evaluated in the ED. Of these, 82 (11%) were deemed to require immediate airway management. Twenty-four of the 82 were excluded because of out-of-hospital traumatic arrest or out-of-hospital intubation, resulting in a study population of 58 patients. Of these 58 patients, 39 had initial rapid sequence intubation using succinylcholine with a 100% success rate. Five comatose patients had successful orotracheal intubation without paralysis, and 2 patients underwent successful emergency tracheostomy. The remaining 12 patients had initial fiberoptic intubation by otolaryngology clinicians, which was unsuccessful in 3 patients. All 3 of these patients were subsequently successfully orotracheally intubated using the rapid sequence intubation technique. Therefore, oral endotracheal intubation was the definitive method of airway management in 47 (81%) of the 58 patients and was successful in all cases.

Conclusion: Rapid sequence intubation was the most commonly performed initial technique by emergency physicians and was safe and effective in all cases attempted. Furthermore,